Training Vectors / Test Vectors (QP Formulation

Use C=1

Training Vectors

x_1	x_2	x_3	x_4
$x_1 = \begin{pmatrix}1\\1\end{pmatrix}$	$x_2 = \begin{pmatrix}2\\2\end{pmatrix}$	$x_3 = \begin{pmatrix}7\\4\end{pmatrix}$	$x_4 = \begin{pmatrix}8\\4\end{pmatrix}$
$\alpha_1 = 1$	$\alpha_2 = 1$	$\alpha_3 = -1$	$\alpha_4 = -1$

Test Vectors

T_1	T_2	T_3
$T_1 = \begin{pmatrix}2\\3\end{pmatrix}$	$T_2 = \begin{pmatrix}4\\5\end{pmatrix}$	$T_3 = \begin{pmatrix}6\\2\end{pmatrix}$

Minimize:

$$\text{Minimize} \quad \mathcal{L}_D(\alpha) = \frac{1}{2}\sum_{j=1}^{N}\sum_{i=1}^{N}\alpha_i\alpha_j y_i y_j [X_i^T X_j + 1]^2 - \sum_{i=1}^{N}\alpha_i$$

= ½ [α1 α2 α3 α4] · Q · [α1; α2; α3; α4] − [α1 + α2 + α3 + α4]

$$\begin{bmatrix} 4.5 & 12.5 & -72 & -84.5 \\ 12.5 & 40.5 & -264.5 & -312.5 \\ -72 & -264.5 & 2178 & 2664.5 \\ -84.5 & -312.5 & 2664.5 & 3280.5 \end{bmatrix}$$

Subject to:

$$\alpha - [\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4]$$

$$\alpha + \alpha \sum_{i=1}^{4} \alpha_i y_i = 0$$

$$\leq 1, \alpha$$

$$\leq 1, \alpha$$

$$\alpha$$

$$_i \leq 1, \alpha$$

$$\geq 0 \quad (i = 1, 2, 3, 4)$$

$$\leq 1$$